

Spin Dynamic Methods

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1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, time is measured in seconds, and magnetic fields $\mathbf{H}$ are expressed in A m⁻¹. The following symbols are used throughout the library.

$\mathbf{m}$ Normalized magnetization, with $\|\mathbf{m}\| = 1$.

$\mathbf{s}_i$ Unit spin direction at site i .

M_s Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$ Effective magnetic field.

$\boldsymbol{\sigma}$ Spin-polarization unit vector.

α Dimensionless Gilbert damping parameter.

γ Positive gyromagnetic ratio, $\gamma = g\mu_B/\hbar$, where g is the Landé g -factor.

γ_0 Positive gyromagnetic factor for an $\mathbf{H}$ -field formulation, $\gamma_0 = \mu_0 |\gamma|$.

θ_{SH} Spin-Hall angle.

J_c Charge-current density.

t_{FM} Ferromagnetic-layer thickness.

β Signed torque ratio, $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$.

$\tau_{\text{DLT}}, \tau_{\text{FLT}}$ Damping-like and field-like torque amplitudes, respectively; both have field units.

e Positive elementary charge.

μ_0 Vacuum permeability.

μ_B Bohr magneton.

$\hbar$ Reduced Planck constant.

k_B Boltzmann constant.

T Absolute temperature.

Physical constants. The following values reproduce `constantfile.m`:

$$\begin{aligned} \gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\ \mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\ \hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}. \end{aligned}$$

Convention warning. Some references define the effective field as $\mathbf{B}_{\text{eff}}$ in tesla and use $|\gamma|$, whereas this library uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} and $\gamma_0 = \mu_0 |\gamma|$. Check this convention before copying an equation into simulation code.

2 Spin Hamiltonians

2.1 Atomistic Spin Hamiltonian

Equation. Following the ASTRO exchange-sign convention, a classical spin Hamiltonian containing exchange, uniaxial anisotropy, Dzyaloshinskii–Moriya interaction, and Zeeman energy is

$$\begin{aligned} \mathcal{H} = & + \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i K_i (\mathbf{s}_i \cdot \hat{\mathbf{e}}_i)^2 \\ & - \frac{1}{2} \sum_{i \neq j} \mathbf{D}_{ij} \cdot (\mathbf{s}_i \times \mathbf{s}_j) - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B}_{\text{ext}}. \end{aligned} \quad (2.1)$$

Definitions. J_{ij} is the exchange-link energy in the ASTRO sign convention: $J_{ij} < 0$ favors parallel alignment, whereas $J_{ij} > 0$ favors antiparallel alignment. K_i is the uniaxial anisotropy energy, $\hat{\mathbf{e}}_i$ is the easy-axis direction, $\mathbf{D}_{ij}$ is the Dzyaloshinskii–Moriya vector, and $\boldsymbol{\mu}_i$ is the magnetic moment at site i .

Assumptions. Each $\mathbf{s}_i$ is a classical unit vector. The pair coefficients satisfy $J_{ji} = J_{ij}$ and $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$. The factors of $1/2$ assume that the pair sums include both ordered pairs (i, j) and (j, i) . If each physical bond is instead included only once, for example in a unique-bond sum $\sum_{\langle i, j \rangle}$, both factors of $1/2$ must be omitted. These are equivalent pair-counting conventions and must not be mixed.

Pair counting and effective field. Let $\mu_i = |\boldsymbol{\mu}_i|$ and define the effective flux density by

$$\mathbf{B}_{\text{eff},i} = -\frac{1}{\mu_i} \frac{\partial \mathcal{H}}{\partial \mathbf{s}_i}. \quad (2.2)$$

For the ordered-pair Hamiltonian above, the exchange and DMI contributions are

$$\mathbf{B}_{\text{ex},i} = -\frac{1}{\mu_i} \sum_{j \neq i} J_{ij} \mathbf{s}_j, \quad \mathbf{B}_{\text{DMI},i} = \frac{1}{\mu_i} \sum_{j \neq i} \mathbf{s}_j \times \mathbf{D}_{ij}. \quad (2.3)$$

The two ordered occurrences of each bond give identical derivatives, which cancel the Hamiltonian factors of $1/2$. Consequently, no extra factor of $1/2$ belongs in either effective-field expression.

Units. All terms in $\mathcal{H}$ have units of energy. Here the external flux density $\mathbf{B}_{\text{ext}}$ is measured in tesla, so no additional factor of μ_0 appears in the Zeeman term.

Source. For the extended classical spin Hamiltonian and Hamiltonian-derivative field convention, see Evans et al. [2].

Implementation notes. The ASTRO implementation evaluates the field, $\mathbf{B}_{\text{eff},i} = -(1/\mu_i) \partial \mathcal{H} / \partial \mathbf{s}_i$, rather than the total energy and includes both forward and backward nearest neighbors. The J_{ij} used above is therefore the coefficient stored in ASTRO's exchange-link arrays: its exchange field is $-(1/\mu_i) \sum_j J_{ij} \mathbf{s}_j$ [5]. No factor of $1/2$ should appear in the exchange or DMI field formulas because differentiation produces two identical bond contributions, cancelling the Hamiltonian factor. ASTRO's DMI component expressions correspond to $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$ and the antisymmetry $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$ used above.

2.2 Macrospin Effective Fields

Field convention. For a fixed-length macrospin with normalized magnetization $\mathbf{m}$, let w denote the magnetic energy density in J m^{-3} . The effective magnetic field and the corresponding effective flux density are [9]

$$\mathbf{H}_{\text{eff}} = -\frac{1}{\mu_0 M_s} \frac{\partial w}{\partial \mathbf{m}}, \quad \mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}. \quad (2.4)$$

Thus, an LLG equation written with $\gamma_0 = \mu_0 |\gamma|$ uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} , whereas one written with $|\gamma|$ uses $\mathbf{B}_{\text{eff}}$ in tesla. The two conventions are physically equivalent but their field variables must not be mixed.

Unless stated otherwise, K_u and A_{12} are energy densities in J m^{-3} . The interfacial K_i in the combination $K_{\text{bulk}} + K_i/t_{\text{FM}}$ is instead in J m^{-2} , and every saturation magnetization is in A m^{-1} . The tensors $\mathbf{N}$ and $\mathbf{K}_{fp}$ are dimensionless.

2.2.1 One-Sublattice Model

Energy density and field decomposition. For uniaxial anisotropy with easy axis $\hat{\mathbf{u}}$, a dimensionless symmetric demagnetization tensor $\mathbf{N}$, an applied field $\mathbf{H}_{\text{ext}}$, and a dipolar field $\mathbf{H}_{\text{dip}}$ produced by another magnetic body, use the uniaxial-anisotropy convention from slides 13–14 of the EE237 Week 5 lecture notes [9] and take

$$w_1 = K_u \left[1 - (\mathbf{m} \cdot \hat{\mathbf{u}})^2 \right] + \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} - \mu_0 M_s \mathbf{m} \cdot (\mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{dip}}). \quad (2.5)$$

External field. The part of the final term in Eq. (2.5) that is proportional to $\mathbf{H}_{\text{ext}}$ is the Zeeman coupling to the applied field. Its derivative contributes $\mathbf{H}_{\text{ext}}$ directly to $\mathbf{H}_{\text{eff}}$ [9].

Uniaxial anisotropy field. Equation (2.4) gives [9]

$$\mathbf{H}_{\text{an}} = \frac{2K_u}{\mu_0 M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}, \quad \mathbf{B}_{\text{an}} = \frac{2K_u}{M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}. \quad (2.6)$$

For $\hat{\mathbf{u}} = \hat{\mathbf{z}}$, the anisotropy term reduces to $K_u(1 - m_z^2) = K_u \sin^2 \theta$, as on slide 13. Near $m_z = 1$, slide 14 therefore gives $H_{\text{an},z} = 2K_u/(\mu_0 M_s)$. For a thin film with bulk and interfacial uniaxial anisotropy, $K_u = K_{\text{bulk}} + K_i/t_{\text{FM}}$.

Demagnetization field. Slide 21 of the same notes writes the demagnetization self-energy as [9]

$$E_{\text{demag}} = -\frac{\mu_0}{2} \int_V \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\text{demag}}(\mathbf{r}) \, d^3r, \quad (2.7)$$

$$w_{\text{demag}} = -\frac{\mu_0}{2} \mathbf{M} \cdot \mathbf{H}_{\text{demag}} = \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} \quad (\text{uniform macrospin}). \quad (2.8)$$

The factor 1/2 avoids double counting the magnet's self-interaction. Equation (2.4) gives

$$\mathbf{H}_{\text{demag}} = -M_s \mathbf{N} \mathbf{m}, \quad \mathbf{B}_{\text{demag}} = -\mu_0 M_s \mathbf{N} \mathbf{m}. \quad (2.9)$$

For principal axes, $\mathbf{N} = \text{diag}(N_x, N_y, N_z)$ and $N_x + N_y + N_z = 1$.

Dipolar field. Let a pinned macrospin have volume V_p , saturation magnetization $M_{s,p}$, and direction $\mathbf{m}_p$, so its magnetic moment is $\boldsymbol{\mu}_p = V_p M_{s,p} \mathbf{m}_p$. In the point-dipole approximation, at displacement $\mathbf{r} = r \hat{\mathbf{r}}$ from the pinned layer to the free layer, slide 22 gives the dipole formula [9]. The slide denotes it by $\mathbf{H}_i$, but its factor μ_0 makes it a flux density in tesla; therefore, in the convention of this document,

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu}_p \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \boldsymbol{\mu}_p] = \mu_0 \mathbf{H}_{\text{dip}}, \quad (2.10)$$

Thermal field. Following Eq. (4) of Zhu et al. [6], write the sampled thermal field as

$$\mathbf{B}_{\text{th}} = \mu_0 \mathbf{H}_{\text{th}} = \mathcal{N}_1(0, u_B) \hat{\mathbf{x}} + \mathcal{N}_2(0, u_B) \hat{\mathbf{y}} + \mathcal{N}_3(0, u_B) \hat{\mathbf{z}}, \quad (2.11)$$

$$u_B = \sqrt{\frac{2\alpha k_B T}{V_{\text{FL}} M_s |\gamma| (1 + \alpha^2) \Delta t}}. \quad (2.12)$$

Here the $\mathcal{N}_i(0, u_B)$ are independent normal random variables with zero mean and standard deviation u_B , V_{FL} is the free-layer volume, and Δt is the duration over which a sampled field is held constant. With $|\gamma|$ in $\text{rad s}^{-1} \text{T}^{-1}$ and M_s in A m^{-1} , u_B is in tesla; the corresponding $\mathbf{H}$ -field standard deviation is $u_H = u_B / \mu_0$.

The source uses independent samples over $\Delta t = 5 \text{ ps}$ [6]. When adopting another integration interval or LLG convention, recompute the noise amplitude consistently rather than reusing that numerical value.

Total field. The total one-sublattice field, with the stochastic field added after taking the energy derivative, is [8]

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{an}} + \mathbf{H}_{\text{demag}} + \mathbf{H}_{\text{dip}} + \mathbf{H}_{\text{th}}. \quad (2.13)$$

Spin-transfer and spin-orbit torques are nonconservative and are not part of the energy derivative in Eq. (2.4).

Implementation mapping. The current one-sublattice `ODE.LLG.integral` implementation stores flux-density quantities in tesla [8]. Its `field_eta.m` evaluates the following expressions, where its input named `Hk` is B_k :

$$\begin{aligned} \mathbf{B}_{\text{an}} &= B_k m_y \hat{\mathbf{y}} \quad (\text{IMA}), & \mathbf{B}_{\text{an}} &= B_k m_z \hat{\mathbf{z}} \quad (\text{PMA}), \\ \mathbf{B}_{\text{demag}} &= -\mu_0 M_s \mathbf{N} \mathbf{m}, & \mathbf{B}_{\text{dip}} &= \mu_0 K_{12} M_s \mathbf{m}_{\text{PL}}, \end{aligned}$$

after converting the input M_s from emu cm^{-3} to A m^{-1} . It sums these with the applied and thermal fields.

2.2.2 Two-Sublattice Collinear Model

Inter-sublattice exchange field. Let $\mathbf{m}_1$ and $\mathbf{m}_2$ be separately normalized sublattice magnetizations with saturation magnetizations $M_{s,1}$ and $M_{s,2}$. The exchange-energy density convention used by the two-sublattice ferrimagnetic macrospin model of Yuan et al. [7] is

$$w_{\text{ex},12} = A_{12} \mathbf{m}_1 \cdot \mathbf{m}_2, \quad A_{12} > 0, \quad (2.14)$$

so $A_{12} > 0$ favors antiparallel alignment. Applying $\mathbf{H}_{\text{ex},i} = -(\mu_0 M_{s,i})^{-1} \partial w_{\text{ex},12} / \partial \mathbf{m}_i$ yields, for $j \neq i$,

$$\mathbf{H}_{\text{ex},i} = -\frac{A_{12}}{\mu_0 M_{s,i}} \mathbf{m}_j, \quad \mathbf{B}_{\text{ex},i} = -\frac{A_{12}}{M_{s,i}} \mathbf{m}_j. \quad (2.15)$$

In that source, $A_{12} = A_{\text{ex}} > 0$. If the exchange energy is instead written $-J_{12} \mathbf{m}_1 \cdot \mathbf{m}_2$, then $\mathbf{H}_{\text{ex},i} = J_{12} \mathbf{m}_j / (\mu_0 M_{s,i})$ and antiferromagnetic coupling corresponds to $J_{12} < 0$. Because each fixed-length sublattice is spatially uniform, an intra-sublattice exchange energy is orientation independent and produces no macrospin torque; spatial exchange stiffness is therefore absent from this model class.

Only the exchange contribution unique to the two-sublattice model is stated here; the shared external, anisotropy, demagnetization, dipolar, and thermal field formulas are defined above and are not repeated. This two-sublattice exchange extension is publication-grounded and is not implemented by the `ODE_LLG_integral` repository.

3 Magnetization Dynamics

LLGS form with field-like torque. This macrospin equation is adapted from Eqs. (1)–(4) of Kong et al., with the atom index removed [3].

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (3.1)$$

Equivalent LL form with field-like torque. The algebraically equivalent form follows the same source convention [3].

$$\begin{aligned} (1 + \alpha^2) \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{H}_{\text{eff}} - \alpha \gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{H}_{\text{eff}}) \\ & + \alpha \gamma \tau_{\text{DLT}} \mathbf{m} \times \boldsymbol{\sigma} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \alpha \gamma \tau_{\text{FLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (3.2)$$

Torque amplitudes. For a ferromagnetic layer [4],

$$\tau_{\text{DLT}} = \frac{\hbar \theta_{\text{SH}} J_c}{2e t_{\text{FM}} M_s}, \quad \tau_{\text{FLT}} = \beta \tau_{\text{DLT}}.$$

Torque-sign convention. In the convention of Eq. (3.1), the sign product of the damping-like and field-like SOT is negative when $\beta > 0$ [1, 3]. Specifically, the scalar prefactors of their respective vector structures are $-\gamma \tau_{\text{DLT}}$ and $+\gamma \tau_{\text{FLT}}$, so their sign product is

$$(-\gamma \tau_{\text{DLT}})(+\gamma \tau_{\text{FLT}}) = -\gamma^2 \beta \tau_{\text{DLT}}^2 < 0 \quad (\beta > 0).$$

Thus, a positive β means that the two SOT terms have oppositely signed scalar prefactors in the LLGS equation, even though the signed amplitudes τ_{DLT} and τ_{FLT} themselves have the same sign.

4 Thermal Stability

4.1 Thermal Stability Factor

Equation. For a single energy barrier E_b , the thermal stability factor is [10]

$$\Delta = \frac{K_u V}{k_B T} \quad (4.1)$$

Units. E_b and $k_B T$ must use the same energy unit; Δ is dimensionless.

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- [9] EE237 Week 5 lecture slides, Magnetic Anisotropy, Magnetostatics, and Zeeman Energy, slides 13–14 and 21–22.
- [10] EE237 Week 10 lecture slides, MRAM Cells, Reliability, and Design Tradeoffs, slide 28.